CS452 - Deep Learning Assignment 1

Name: Minahil Tariq
FastID: 22i-2012

1. Introduction

This assignment implements a multi-task CNN pipeline to (1) classify facial expressions into 8 categories and (2) predict continuous valence and arousal values in [-1, +1]. Two modern CNN backbones were evaluated using transfer learning: **MobileNetV2** and **EfficientNetB0**.

2. Dataset Details

The dataset contains facial images along with annotations for expression (classification), valence and arousal (regression), and additional indices.

- Images: 3999 total
- Annotation files: expression, valence, arousal, index

Dataset Split:
- Training: 3199 images
- Validation: 800 images

3. Model Architectures

Two modern CNN backbones were implemented for multitask learning:

1. MobileNetV2
   - Pretrained on ImageNet
   - Lightweight and efficient
   - Trained for 8 epochs

2. EfficientNetB0
   - Pretrained on ImageNet
   - Parameter-efficient
   - Trained for 8 epochs

4. Training Settings

- Image size: 128 × 128
- Batch size: 32
- Optimizer: Adam (learning rate = 1e-4)
- Losses: CrossEntropy (classification) + MSE (regression)
- Epochs: 8
- Hardware: Google Colab GPU

5. Performance Comparison

Model	Accuracy	F1	Kappa	AUC	AP	RMSE	Correlation	SAGR	CCC
MobileNetV2	0.3088	0.298	0.21	0.709	0.285	[0.520, 0.439]	[0.189, 0.156]	[0.658, 0.744]	[0.177, 0.144]
EfficientNetB0	0.144	0.059	0.021	0.533	0.137	[0.467, 0.385]	[0.001, 0.026]	[0.694, 0.776]	[0.000, 0.000]

6. Discussion & Comparison

- MobileNetV2 gave the best classification accuracy in this quick experiment and exhibited non-trivial AUC and kappa values — showing the network captured useful features even at 128×128 and with few epochs.
- EfficientNetB0 did not learn useful class separation in 3 epochs (accuracy stayed near chance). This is unsurprising for a height-parameterized model when not fine-tuned; increasing epochs and using model-specific preprocessing could help.
- Time vs accuracy: MobileNetV2 is lighter and faster on GPU than EfficientNetB0; for the assignment, MobileNetV2 provided the best balance in our quick run.

7. Training Curves

Training and validation accuracy curves were plotted for each backbone to analyze convergence.
MobileNetV2 showed gradual improvement in accuracy over epochs, while EfficientNetB0 remained near chance.

8. Observations & Conclusion

1. MobileNetV2 achieved ~31% accuracy, showing some learning above random baseline.
2. EfficientNetB0 remained close to random guessing, indicating underfitting in limited epochs.
3. Regression results showed modest RMSE but very low correlation and CCC.
4. Overall, MobileNetV2 is a better baseline for multitask FER (Facial Expression Recognition) on this dataset.